Vector addition

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Add displacement vectors and use a model to explain where an object ends up.

SUCCESS CRITERIA - I can...

1. Draw and label arrows tip to tail; keep their length and direction.
2. Draw the resultant from the original start to the final endpoint.
3. State net east-west and north-south motion with units and direction.

Core: teacher selects from 1-10. **Stretch:** 11-12 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “vector” and “component” to explain your model or result. Sentence starters are optional.

vector: a quantity with size and direction

component: the part of a vector along one chosen axis

resultant: one vector equal to the sum of other vectors

01 Warm-up: bring back an earlier idea

A cart moves from 0 m to 8 m east in 4 s at a steady pace.
Find its speed.
Draw a velocity arrow showing its direction.
Explain what the arrow tells you that speed alone does not.

Word help: velocity = rate of position change, with direction

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

02 Read and explain a worked model

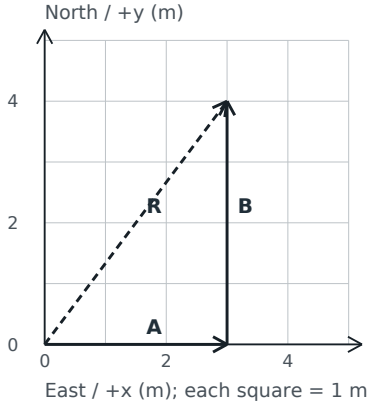
Reference: $R_x = A_x + B_x$; $R_y = A_y + B_y$.
Use east = +x and north = +y. Add signed components; draw R from the original start to the final endpoint.

Worked reasoning: Add east-west and north-south motion separately: R_x = 3 m and R_y = 4 m. The route length is 3 + 4 = 7 m. The supplied straight-line length is 5 m.

Your turn: A walker goes 3 m east, then 4 m north. On the model, circle the original start and final endpoint.
Explain why the distance is 7 m while the displacement magnitude is 5 m.

Word help: magnitude = size of a quantity without its direction

Sentence starter: My claim is … . The diagram/data show …, so … .



Marzano check - circle: ① ② ③

Vector addition

antocciphysics

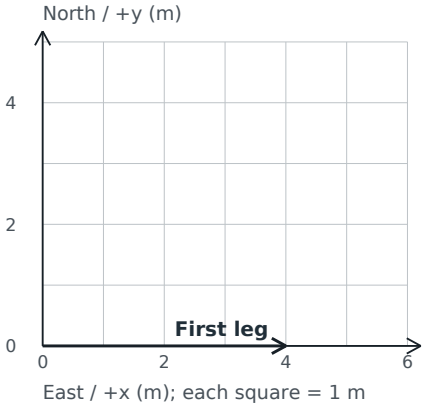
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Finish the model

Walk 4 m east, then 3 m north. The first arrow is drawn. Add the second arrow from its tip and a dashed resultant from the original start.

Sentence starter: The ... in my model represents ... . Its label shows ... .

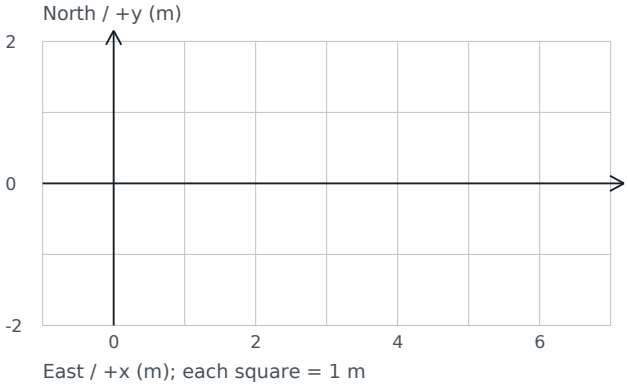


Marzano check - circle: ① ② ③

04 A trip that doubles back

Walk 6 m east, then 4 m west.
Draw and label the route and resultant.
Give distance and displacement.
Explain which length follows the path.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

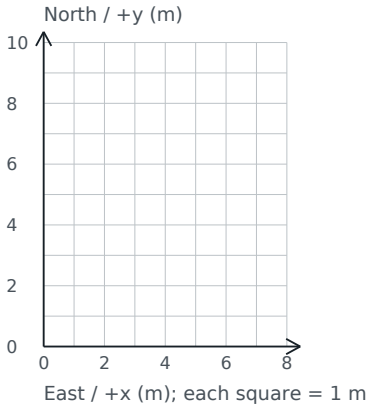


Marzano check - circle: ① ② ③

05 Use a new scale

Walk 6 m east, then 8 m north.
Draw both arrows and the resultant.
Give distance and displacement. The straight-line length is supplied: 10 m.

Sentence starter: The ... in my model represents ... . Its label shows ... .

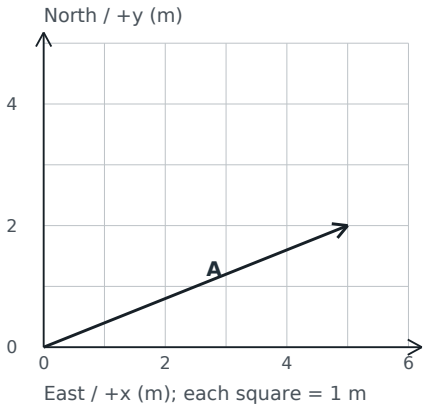


Marzano check - circle: ① ② ③

06 Add two diagonal arrows

A: 5 m east and 2 m north. B: 3 m west and 1 m north. A is drawn. Add B and R.
State each net component and the general compass direction.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

Vector addition

antocciphysics

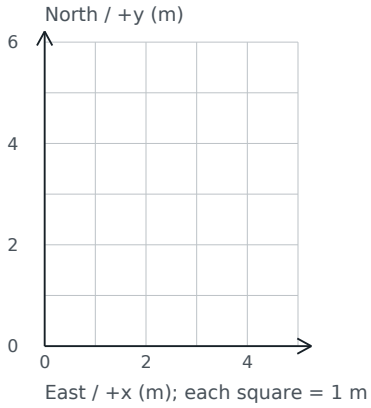
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Predict, draw, check

Walk 3 m east, 4 m north, then 3 m west.
Predict what cancels.
Draw the route and R.
Give distance and displacement.
Check your prediction against the model.

Sentence starter: I predict … . My model supports this because … .

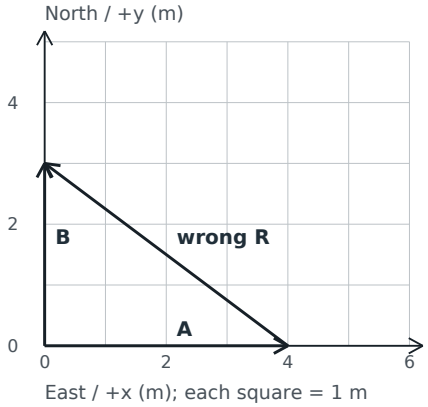


Marzano check - circle: 1 2 3

08 Find and repair the mistake

Sam draws the two displacement arrows from the same start, then connects their tips and calls that arrow the resultant. The model shows his mistake.
Draw the correct tip-to-tail route beside it.
Explain where the correct resultant must begin and end.

Sentence starter: The step I would change is … . The evidence is … .



Marzano check - circle: 1 2 3

09 One trip, three questions

A student walks 6 m east in 6 s, then 8 m north in 8 s. The direct start-to-finish length is 10 m. For each answer, decide whether to use path length or the start-to-finish arrow, then solve:
(a) Distance walked? (b) Displacement?
(c) Average speed for the whole trip?
Label the model with the given positions and any prediction.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I predict … . My model supports this because … .

Route



Position (m)

Marzano check - circle: 1 2 3

Vector addition

antocciphysics

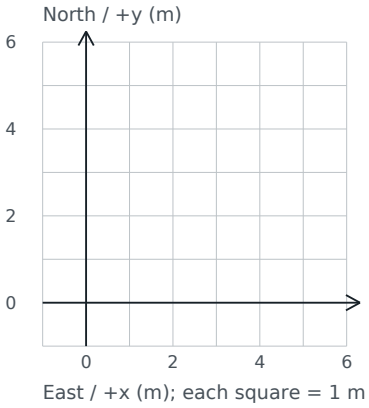
Continue down each column.

Sentence starters are optional. You may plan in any language.

10 Exit check - work independently

A robot makes two moves. First: 4 m east and 1 m north. Second: 1 m west and 3 m north.
Draw both moves tip-to-tail and add the resultant.
State its two components and general compass direction.
Explain one way to check the result.

Sentence starter: My claim is … . The diagram/data show …, so … .

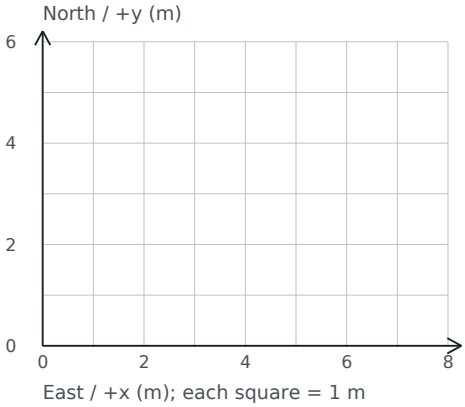


Marzano check - circle: ① ② ③

11 STRETCH: Design two routes

Finish 4 m east and 3 m north of your start.
Design one route of length 7 m and another of length 11 m.
Use only north, south, east or west legs.
Label all lengths and draw each resultant.

Sentence starter: My design shows … . It meets the requirement by … .



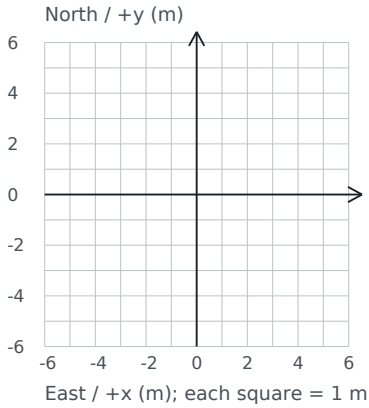
Marzano check - circle: ① ② ③

12 STRETCH: Explore the limits

Point two 5 m displacement arrows in any directions.
Draw the smallest and largest possible resultants, then one between those extremes.
Estimate the intermediate resultant and explain why your limits make sense.

Word help: estimate = find an approximate value

Sentence starter: My claim is … . The diagram/data show …, so … .



Marzano check - circle: ① ② ③